

Maurer–Cartan Obstruction Towers in Completed Totalizations

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Abstract

Local deformation conditions often appear as projections of one nonlinear equation. Deforming two algebras, for example, leaves a separate obstruction to deforming a map between them. We compute that obstruction and exhibit a flat algebra deformation whose augmentation cannot lift. We then construct the common carrier for a semicosimplicial diagram as a completed Thom–Whitney differential graded Lie algebra and prove its full obstruction tower. Strict completeness identifies compatible Maurer–Cartan representatives with a total solution. Gauge classes obey a different limit theorem: a generally nonabelian $\varprojlim^1$ of derived stabilizers measures their ambiguity. A second Milnor sequence controls compatible primitives in inverse systems of complexes. We also give the filtered L_∞ form, the convolution carrier for comparison morphisms, and sharp examples showing why completeness, separation, additive filtration, flatness, and totalization hypotheses cannot be omitted.

1 A curvature invisible in every projected equation

An operation on a boundary must remain compatible with boundary composition when the bulk and boundary theories deform. The two deformed theories therefore do not determine a deformed action. Even ordinary associative algebras exhibit this distinction: a flat deformation can destroy the augmentation that specifies its one-dimensional representation. The calculation in Section 5.1 identifies the additional equation and its obstruction class. The question for a system of local operations is how to construct these coupled equations, solve their successive obstructions, and assemble compatible solutions.

Let $\mathbb{k}$ be a field of characteristic zero. We use cohomological degrees: d has degree 1, and the bracket satisfies

$$[u, v] = -(-1)^{|u||v|}[v, u], \quad d[u, v] = [du, v] + (-1)^{|u|}[u, dv].$$

For an element a of degree 1, write

$$\text{Curv}(a) = da + \frac{1}{2}[a, a].$$

Example 1.1 (The first missing equation). Let

$$\mathfrak{g}^1 = \mathbb{k}x, \quad \mathfrak{g}^2 = \mathbb{k}z, \quad d = 0, \quad [x, x] = 2z,$$

with z central. Give $\mathfrak{g}$ the filtration

$$F^1\mathfrak{g} = \mathfrak{g}, \quad F^2\mathfrak{g} = \mathbb{k}z, \quad F^3\mathfrak{g} = 0.$$

Then

$$\text{Curv}(x) = z.$$

Thus x is Maurer–Cartan modulo $F^2\mathfrak{g}$, but it does not lift modulo $F^3\mathfrak{g}$. Moreover, the projection

$$p : \mathfrak{g} \longrightarrow \mathfrak{g}/F^2\mathfrak{g}$$

sends x to a Maurer–Cartan element and sends its curvature to zero. A family of such projections can therefore erase a nonzero common obstruction. The obstruction is not a list of projected classes: here it is the single class

$$[z] \in H^2(F^2\mathfrak{g}/F^3\mathfrak{g}) = \mathbb{k}z.$$

The missing class belongs to the common complex before projection. A filtration isolates its first nonzero term and makes the correction equation linear. Local corrections must then satisfy the overlap equations in a common totalization. These are the two constructions developed below.

2 The finite obstruction step

Definition 2.1. An *additively filtered dg Lie algebra* is an uncurved dg Lie algebra $(\mathfrak{g}, d, [-, -])$ with a descending filtration by subcomplexes

$$\mathfrak{g} = F^1\mathfrak{g} \supset F^2\mathfrak{g} \supset \cdots$$

such that

$$[F^p\mathfrak{g}, F^q\mathfrak{g}] \subset F^{p+q}\mathfrak{g}.$$

It is *strictly complete and separated* when the canonical map

$$\mathfrak{g} \longrightarrow \varprojlim_r \mathfrak{g}/F^r\mathfrak{g} \quad (2.1)$$

is an isomorphism of graded vector spaces, hence of complexes. We abbreviate

$$\mathfrak{g}_r = \mathfrak{g}/F^{r+1}\mathfrak{g}, \quad A_r = F^r\mathfrak{g}/F^{r+1}\mathfrak{g}.$$

The extension

$$0 \longrightarrow A_r \longrightarrow \mathfrak{g}_r \longrightarrow \mathfrak{g}_{r-1} \longrightarrow 0 \quad (2.2)$$

is central and has square-zero bracket on its kernel. This elementary fact is the entire linearization mechanism.

Theorem 2.2 (One-stage obstruction and relative lifts). *Let $r \geq 1$, and let $a_{r-1} \in \text{MC}(\mathfrak{g}_{r-1})$. For any graded lift $\tilde{a} \in \mathfrak{g}_r^1$, its curvature belongs to A_r^2 and determines a lift-independent class*

$$o_r(a_{r-1}) = [\text{Curv}(\tilde{a})] \in H^2(A_r, d). \quad (2.3)$$

The element a_{r-1} lifts to $\text{MC}(\mathfrak{g}_r)$ if and only if $o_r(a_{r-1}) = 0$.

If lifts exist, their set is a torsor under $Z^1(A_r, d)$. After quotienting only by gauge transformations that reduce to the identity on $\mathfrak{g}_{r-1}$, the set of lifts is a torsor under $H^1(A_r, d)$, and the relative stabilizer is $Z^0(A_r, d)$.

Proof. The twisted operator $d_{\tilde{a}} = d + [\tilde{a}, -]$ induces d on A_r , because

$$[F^1\mathfrak{g}, F^r\mathfrak{g}] \subset F^{r+1}\mathfrak{g}.$$

The Bianchi identity

$$d_{\tilde{a}} \text{Curv}(\tilde{a}) = 0 \quad (2.4)$$

therefore makes the curvature a cocycle in A_r . A second lift has the form $\tilde{a} + \beta$ with $\beta \in A_r^1$, and centrality in (2.2) gives the exact identity in $\mathfrak{g}_r$

$$\text{Curv}(\tilde{a} + \beta) = \text{Curv}(\tilde{a}) + d\beta. \quad (2.5)$$

This proves independence and the lifting criterion. Once a lift is fixed, all others differ from it by $Z^1(A_r)$.

The relative gauge group is $\exp(A_r^0)$. Since A_r is central in $\mathfrak{g}_r$, its action is

$$\tilde{a} \mapsto \tilde{a} - d\xi, \quad \xi \in A_r^0.$$

Taking relative orbits and stabilizers gives $H^1(A_r)$ and $Z^0(A_r)$. $\square$

Quotienting by all gauge transformations lying above a gauge orbit in $\mathfrak{g}_{r-1}$ is different: the stabilizer of the base element can act on $H^1(A_r)$. The torsor assertion in Theorem 2.2 is deliberately relative.

For a strictly complete filtered dg Lie algebra the Baker–Campbell–Hausdorff series converges on $F^1\mathfrak{g}^0$. With our signs the gauge action is

$$e^\xi \cdot a = e^{\text{ad}\xi} a - \frac{e^{\text{ad}\xi} - 1}{\text{ad}\xi}(d\xi) = a - \sum_{n \geq 0} \frac{(\text{ad}\xi)^n}{(n+1)!} d_a \xi. \quad (2.6)$$

Its infinitesimal vector is $-d_a \xi$, and

$$\text{Curv}(e^\xi \cdot a) = e^{\text{ad}\xi} \text{Curv}(a). \quad (2.7)$$

The series converge because every additional bracket with $\xi \in F^1$ moves a term at least one step deeper. Equation (2.7) also shows directly that (2.3) is gauge invariant.

Corollary 2.3 (The recursive curvature equation). *Suppose that the filtration is split by a complete weight decomposition*

$$\mathfrak{g} = \prod_{p \geq 1} \mathfrak{g}^{(p)}, \quad F^r \mathfrak{g} = \prod_{p \geq r} \mathfrak{g}^{(p)},$$

and that $d\mathfrak{g}^{(p)} \subset \mathfrak{g}^{(p)}$ and $[\mathfrak{g}^{(i)}, \mathfrak{g}^{(j)}] \subset \mathfrak{g}^{(i+j)}$. Write

$$a = \sum_{p \geq 1} a^{(p)}, \quad a^{(p)} \in \mathfrak{g}^{(p),1}.$$

Then its weight- r Maurer–Cartan equation is

$$da^{(r)} + \frac{1}{2} \sum_{i+j=r} [a^{(i)}, a^{(j)}] = 0. \quad (2.8)$$

Consequently the first unknown correction is governed by the class of the known quadratic term in $H^2(\text{gr}_F^r \mathfrak{g}, d)$.

3 The completed Thom–Whitney carrier

The finite obstruction step applies on each open set. To glue its solutions, the overlap equations must belong to a differential graded Lie algebra as well. Polynomial differential forms on simplices retain the homotopies between restrictions and their compatibility on multiple overlaps.

Let $\mathfrak{g}^\bullet$ be a semicosimplicial additively filtered dg Lie algebra. Thus there are filtration-preserving coface maps

$$\partial^i : \mathfrak{g}^{q-1} \longrightarrow \mathfrak{g}^q \quad (0 \leq i \leq q)$$

satisfying the semicosimplicial identities, and each $\mathfrak{g}^q$ is strictly complete and separated. Put

$$\Omega_q = \frac{\mathbb{k}[t_0, \dots, t_q, dt_0, \dots, dt_q]}{(\sum_i t_i - 1, \sum_i dt_i)}, \quad |t_i| = 0, \quad |dt_i| = 1,$$

with $d_{\text{dR}} t_i = dt_i$. The pullback to the i th face is denoted $\delta_i^* : \Omega_q \rightarrow \Omega_{q-1}$. Define the completed tensor product by

$$\Omega_q \widehat{\otimes} \mathfrak{g}^q = \varprojlim_r (\Omega_q \otimes (\mathfrak{g}^q / F^r \mathfrak{g}^q)). \quad (3.1)$$

Definition 3.1 (Completed Thom–Whitney end). The completed Thom–Whitney totalization is

$$\widehat{\text{TW}}(\mathfrak{g}^\bullet) = \left\{ (x_q)_{q \geq 0} \in \prod_{q \geq 0} \Omega_q \widehat{\otimes} \mathfrak{g}^q \mid (\delta_i^* \otimes 1)x_q = (1 \otimes \partial^i)x_{q-1} \right\}. \quad (3.2)$$

On homogeneous tensors its differential and bracket are

$$D(\omega \otimes x) = d_{\text{dR}} \omega \otimes x + (-1)^{|\omega|} \omega \otimes dx, \quad (3.3)$$

$$[\omega \otimes x, \eta \otimes y] = (-1)^{|x||\eta|} \omega \eta \otimes [x, y]. \quad (3.4)$$

Its filtration is componentwise:

$$F^r \widehat{\text{TW}} = \widehat{\text{TW}} \cap \prod_q \ker(\Omega_q \widehat{\otimes} \mathfrak{g}^q \longrightarrow \Omega_q \otimes (\mathfrak{g}^q / F^r \mathfrak{g}^q)). \quad (3.5)$$

Example 3.2 (The tensor-product sign). Let $\mathfrak{a}$ be abelian with $x \in \mathfrak{a}^0$, $y \in \mathfrak{a}^1$, $dx = y \neq 0$, and $dy = 0$. If the second summand in (3.3) were given a plus sign independently of $|\omega|$, then on $\Omega_1 \otimes \mathfrak{a}$ the resulting operator D_+ would satisfy

$$D_+^2(t_0 \otimes x) = 2 dt_0 \otimes y \neq 0.$$

Thus the factor $(-1)^{|\omega|}$ is necessary even before imposing the face equations.

The signs in (3.3)–(3.4) are forced by the tensor product convention. In particular, replacing (3.2) by an ordinary normalized Čech total complex and appending a symbol for the Čech differential does not produce a dg Lie bracket. The normalized total complex instead receives transferred L_∞ operations from the Thom–Whitney model [2].

Proposition 3.3 (Completeness of the total carrier). *The formulas above make $T = \widehat{\text{TW}}(\mathfrak{g}^\bullet)$ a strictly complete and separated additively filtered dg Lie algebra. For*

$$T_r = \text{TW}(\mathfrak{g}^\bullet / F^r \mathfrak{g}^\bullet),$$

the canonical map $T/F^rT \rightarrow T_r$ is injective. Define

$$I_r = \text{im}(T \rightarrow T_r).$$

Thus I_r consists exactly of the finite-stage elements that admit one coherent lift through all deeper stages. One has

$$T \cong \varprojlim_r T_r \cong \varprojlim_r I_r. \quad (3.6)$$

If every projection $T \rightarrow T_r$ is surjective, then $I_r = T_r$ and $T/F^rT \cong T_r$.

Proof. The product in (3.2) is complete. The face equations cut out the kernel of a continuous linear map to a separated product, hence a closed complete subspace. Equations (3.3) and (3.4) preserve the face equations. Their componentwise form gives

$$D(F^rT) \subset F^rT, \quad [F^pT, F^qT] \subset F^{p+q}T.$$

Inverse limits commute with products and equalizers, so the definition also gives $T \cong \varprojlim_r T_r$. The kernel of $T \rightarrow T_r$ is exactly F^rT , proving injectivity of the induced map and identifying its image with I_r . A compatible sequence in the I_r is, component by component, a compatible sequence in every local quotient; completeness gives an element of the ambient product, and separatedness makes all face equations hold. This proves (3.6). The last assertion is immediate. $\square$

Levelwise surjectivity of $\mathfrak{g}^q/F^{r+1} \rightarrow \mathfrak{g}^q/F^r$ does not by itself imply surjectivity after taking the end (3.2). Proposition 3.3 isolates the extra strictness needed to identify every algebraic finite totalization with a quotient of the completed one.

Theorem 3.4 (Filtered and totalized obstruction tower). *Let $T = \widehat{\text{TW}}(\mathfrak{g}^\bullet)$. For every $r \geq 1$ and every $a_{r-1} \in \text{MC}(T/F^rT)$ there is a canonical class*

$$o_r(a_{r-1}) \in H^2(F^rT/F^{r+1}T, D) \quad (3.7)$$

whose vanishing is equivalent to a lift in $\text{MC}(T/F^{r+1}T)$. When lifts exist, the relative lift and relative gauge-orbit spaces are torsors under $Z^1(F^rT/F^{r+1}T)$ and $H^1(F^rT/F^{r+1}T)$, respectively.

A compatible sequence

$$a_r \in \text{MC}(T/F^{r+1}T), \quad a_r \equiv a_{r-1} \pmod{F^rT}, \quad (3.8)$$

determines a unique element $a \in \text{MC}(T)$. If the strictness condition $T/F^rT \cong T_r$ holds, the same assertions may be read directly in the finite Thom–Whitney totalizations T_r .

Proof. The finite assertions are Theorem 2.2 applied to T . Proposition 3.3 gives a unique degree-one element $a \in T$ from (3.8). Its curvature is zero in every quotient, so

$$\text{Curv}(a) \in \bigcap_r F^rT^2 = 0.$$

$\square$

A natural transformation of semicosimplicial filtered dg Lie algebras induces a strict filtered map of completed Thom–Whitney carriers. More generally, suppose

$$P_i : T \rightarrow R_i \quad (i \in J)$$

are filtered dg Lie maps, let $a_{r-1} \in \text{MC}(T/F^r T)$, and write

$$(P_i)_{r-1} : T/F^r T \longrightarrow R_i/F^r R_i$$

for the induced map. Naturality of curvature gives

$$o_r((P_i)_{r-1}(a_{r-1})) = H^2(\text{gr}_F^r P_i) o_r(a_{r-1}). \quad (3.9)$$

Hence the vanishing of all projected classes forces the total class to vanish when

$$H^2(\text{gr}_F^r T) \longrightarrow \prod_{i \in J} H^2(\text{gr}_F^r R_i) \quad (3.10)$$

is injective. Without this hypothesis, or a direct construction assembling the projected primitives into a primitive in T , the converse fails by Example 1.1.

4 Complete filtered L_∞ algebras

Transferring the totalization to a smaller complex generally introduces higher brackets. The same obstruction argument applies when their filtration makes the curvature series converge. We use the shifted symmetric convention to state this extension. A flat shifted L_∞ algebra is a graded vector space L with graded symmetric maps

$$q_n : S^n L \longrightarrow L, \quad |q_n| = 1, \quad n \geq 1, \quad (4.1)$$

whose induced coderivation on the reduced symmetric coalgebra squares to zero. There is no nullary bracket. If $L = \mathfrak{g}[1]$, degree 0 in L corresponds to degree 1 in the unshifted dg Lie algebra $\mathfrak{g}$; consequently $H^1(L, q_1)$ corresponds to $H^2(\mathfrak{g}, d)$.

Assume

$$L = F^1 L \supset F^2 L \supset \cdots, \quad q_n(F^{p_1} L, \dots, F^{p_n} L) \subset F^{p_1 + \cdots + p_n} L, \quad L \cong \varprojlim_r L/F^r L. \quad (4.2)$$

For $a \in L^0$, define

$$R(a) = \sum_{n \geq 1} \frac{1}{n!} q_n(a^n), \quad (4.3)$$

$$q_m^a(x_1, \dots, x_m) = \sum_{k \geq 0} \frac{1}{k!} q_{m+k}(a^k, x_1, \dots, x_m). \quad (4.4)$$

The n th summand in (4.3) lies in $F^n L$, so the series converges. Additivity similarly gives convergence of every twisted bracket. The L_∞ identities imply

$$q_1^a R(a) = 0, \quad (4.5)$$

$$R(a+b) = R(a) + \sum_{m \geq 1} \frac{1}{m!} q_m^a(b^m). \quad (4.6)$$

These formulas and their filtered convergence are developed in [1, 6].

Proposition 4.1 (The L_∞ obstruction). *If $R(a) \in F^r L^1$, then its image defines*

$$o_r(a) = [R(a)] \in H^1(F^r L/F^{r+1} L, q_1). \quad (4.7)$$

There is $b \in F^r L^0$ with $R(a+b) \in F^{r+1} L$ if and only if $o_r(a) = 0$.

Proof. On F^r/F^{r+1} the operator q_1^a reduces to q_1 , so (4.5) makes $R(a)$ a q_1 -cocycle. For $b \in F^r L^0$, every term of (4.6) except $q_1 b$ lies in $F^{r+1}L$. Thus

$$R(a+b) \equiv R(a) + q_1 b \pmod{F^{r+1}L}.$$

□

Without completeness the infinite expression (4.3) need not exist in L . The finite statement remains valid inside $L/F^{r+1}L$, where additivity makes all but finitely many summands vanish.

A filtered shifted L_∞ morphism $U : L \rightsquigarrow M$ has degree-zero Taylor coefficients

$$U_n : S^n L \longrightarrow M, \quad U_n(F^{p_1}, \dots, F^{p_n}) \subset F^{p_1+\dots+p_n} M.$$

It defines

$$U_*(a) = \sum_{n \geq 1} \frac{1}{n!} U_n(a^n),$$

and satisfies the off-shell curvature identity

$$R_M(U_*(a)) = \sum_{m \geq 0} \frac{1}{m!} U_{m+1}(a^m, R_L(a)). \quad (4.8)$$

Consequently

$$o_r(U_* a) = H^1(\mathrm{gr}_F^r U_1) o_r(a). \quad (4.9)$$

For a non-strict morphism the right side of (4.8) is not merely $U_1 R_L(a)$.

There is no canonical exponential gauge group for a general L_∞ algebra. Its intrinsic equivalence relation is defined by the connected components of the completed simplicial Maurer–Cartan set

$$\mathrm{MC}_\bullet(L) = \mathrm{MC}(L \hat{\otimes} \Omega_{\mathrm{poly}}(\Delta^\bullet)), \quad L \hat{\otimes} \Omega_n = \varprojlim_r (L/F^r L) \otimes \Omega_n. \quad (4.10)$$

For dg Lie algebras this recovers the homotopy theory of (2.6); for nilpotent quotients it is the Deligne–Getzler–Hinich construction [3].

5 Comparison morphisms and colored operations

The Taylor coefficients of a comparison morphism themselves form a complete filtered carrier. Let L and M be flat shifted L_∞ algebras, let $C(L) = \overline{S}^c(L)$ carry the coderivation Q_L , and set

$$\mathcal{H}(L, M) = \mathrm{Hom}(C(L), M) = \prod_{n \geq 1} \mathrm{Hom}(S^n L, M). \quad (5.1)$$

Continuous homomorphisms and completed symmetric powers are understood when L and M have additional complete linear topologies. The arity filtration is

$$F^p \mathcal{H}(L, M) = \prod_{n \geq p} \mathrm{Hom}(S^n L, M). \quad (5.2)$$

If f is homogeneous, the unary bracket is

$$q_1^{\mathcal{H}}(f) = q_1^M \circ f - (-1)^{|f|} f \circ Q_L, \quad (5.3)$$

and for $m \geq 2$ the value $q_m^{\mathcal{H}}(h_1, \dots, h_m)$ is obtained by applying the m -fold reduced coproduct of $C(L)$, then $h_1 \otimes \dots \otimes h_m$ with Koszul symmetrization, and finally q_m^M . Thus

$$q_m^{\mathcal{H}}(F^{p_1}, \dots, F^{p_m}) \subset F^{p_1 + \dots + p_m} \mathcal{H}. \quad (5.4)$$

The L_∞ identities follow from $Q_L^2 = Q_M^2 = 0$. A degree-zero element $f = (f_1, f_2, \dots)$ is Maurer–Cartan precisely when the coalgebra map determined by its Taylor coefficients obeys

$$Q_M F = F Q_L; \quad (5.5)$$

that is, precisely when $F : L \rightsquigarrow M$ is an L_∞ morphism. This convolution construction is treated systematically in [5].

A chain map alone does not suffice. For example, let L be the abelian dg Lie algebra on a degree-one element x , and let M have degree-one element y , degree-two central element z , zero differential, and $[y, y] = 2z$. The linear chain map $x \mapsto y$ sends a Maurer–Cartan element to an element of curvature z . The missing bracket identity is one of the higher equations in (5.5).

There is a parallel carrier for colored algebraic structures. Let $\mathcal{C}$ be a reduced conilpotent coaugmented two-colored dg cooperad with coaugmentation coideal $\bar{\mathcal{C}}$. Reduced means that all arity-zero terms vanish and that $\bar{\mathcal{C}}$ has no arity-one term. For a two-colored cochain complex $V = (V_c, V_o)$, define

$$\text{End}_V(c_1, \dots, c_n; c) = \text{Hom}(V_{c_1} \otimes \dots \otimes V_{c_n}, V_c).$$

The convolution dg Lie algebra is

$$\text{Conv}(\bar{\mathcal{C}}, \text{End}_V) = \prod_{\vec{c}; c} \text{Hom}_{\Sigma_{\vec{c}}}(\bar{\mathcal{C}}(\vec{c}; c), \text{End}_V(\vec{c}; c)). \quad (5.6)$$

For homogeneous f its differential is

$$\partial f = d_{\text{End}} f - (-1)^{|f|} f d_{\mathcal{C}}. \quad (5.7)$$

If $\Delta_{(1)}$ is the infinitesimal cooperadic decomposition and $\gamma_{(1)}$ is infinitesimal composition in End_V , put

$$f \star g = \gamma_{(1)}(f \circ_{(1)} g) \Delta_{(1)}, \quad [f, g] = f \star g - (-1)^{|f||g|} g \star f. \quad (5.8)$$

All permutation and insertion signs in (5.8) are the Koszul signs of the colored symmetric sequences.

There is also a precise filtered topological version. Suppose each V_c has a complete separated filtration $G^0 V_c = V_c \supset G^1 V_c \supset \dots$ preserved by its differential. Give an algebraic tensor product the total filtration

$$G^r(V_{c_1} \otimes \dots \otimes V_{c_n}) = \sum_{\substack{r_1, \dots, r_n \geq 0 \\ r_1 + \dots + r_n \geq r}} G^{r_1} V_{c_1} \otimes \dots \otimes G^{r_n} V_{c_n}$$

and define its completed tensor product as the inverse limit of the quotients by G^r . Replacing Hom above by continuous linear maps gives a topological endomorphism operad provided that substitution is continuous; this continuity and the associativity of the completed tensor product are explicit hypotheses. Under them, (5.7)–(5.8) extend continuously.

Suppose $\mathcal{C} = \bigoplus_{w \geq 1} \mathcal{C}^{(w)}$, its differential preserves weight, and $\Delta_{(1)}$ splits weight additively. Maps vanishing on weights below p form a complete product filtration W^p , and

$$[W^p, W^q] \subset W^{p+q}. \quad (5.9)$$

For an arity filtration under operadic insertion the correct positive weight is arity minus one, since

$$(m-1) + (n-1) = (m+n-1) - 1.$$

Thus

$$W^p = \{\text{operations of total arity at least } p+1\} \quad (5.10)$$

is additive, whereas filtration by arity at least p satisfies only $[F^p, F^q] \subset F^{p+q-1}$. Unary structure must be fixed and absorbed into the differential, or into a twist by a fixed base Maurer–Cartan element, before the positive-weight deformation variable is pronilpotent.

Here is the cobar convention compatible with our cohomological grading. Put $(E[k])^n = E^{n+k}$, and let

$$\Omega_{\text{coh}}\mathcal{C} = (\mathcal{F}_{\text{op}}(\overline{\mathcal{C}}[-1]), d_{\Omega}). \quad (5.11)$$

The inclusion of generators $\kappa : \overline{\mathcal{C}} \rightarrow \Omega_{\text{coh}}\mathcal{C}$ has degree 1. The cobar differential is the unique operadic derivation determined by the universal twisting equation; writing $\gamma_{(1)}^{\mathcal{F}}$ for infinitesimal composition in the free colored operad, this equation is

$$d_{\Omega}\kappa(c) = -\kappa(d_{\mathcal{C}}c) - \gamma_{(1)}^{\mathcal{F}}(\kappa \circ_{(1)} \kappa)\Delta_{(1)}(c). \quad (5.12)$$

The cooperad identities give $d_{\Omega}^2 = 0$. A dg operad map $\Phi : \Omega_{\text{coh}}\mathcal{C} \rightarrow \text{End}_V$ is determined by the degree-one element $\theta = \Phi\kappa$ and commutes with differentials exactly when

$$\partial\theta + \theta \star \theta = \partial\theta + \frac{1}{2}[\theta, \theta] = 0. \quad (5.13)$$

Thus Maurer–Cartan elements of (5.6) are precisely $\Omega_{\text{coh}}\mathcal{C}$ -algebra structures on V . A specified quasi-free resolution $\Omega_{\text{coh}}\mathcal{C} \rightarrow \mathcal{P}$ of a reduced two-colored dg operad $\mathcal{P}$ therefore supplies a carrier for homotopy $\mathcal{P}$ -algebra structures. An operad cannot be placed in the cooperadic entry of (5.6) without such a bar or cobar construction. Fixed endpoints may be imposed by twisting the full carrier by their base structure and taking the kernel of a strict filtered endpoint map. If the endpoint map exists only up to coherent homotopy, its specified L_{∞} homotopy fibre is required; an unadorned mapping cone is only a complex and has no automatic nonlinear brackets.

Assume now that the weight filtration is split and that every bracket has output weight equal to the sum of its input weights. Write a positive-weight deformation as

$$\theta = \sum_{p \geq 1} \theta^{(p)}.$$

For a complete filtered L_{∞} carrier, the exact weight- r equation is

$$q_1\theta^{(r)} + \sum_{n=2}^r \frac{1}{n!} \sum_{i_1+\dots+i_n=r} q_n(\theta^{(i_1)}, \dots, \theta^{(i_n)}) = 0. \quad (5.14)$$

In the dg Lie case, a closed binary term $\theta^{(1)}$ has the first curvature

$$\text{Curv}(\theta^{(1)}) \equiv \frac{1}{2}[\theta^{(1)}, \theta^{(1)}] \pmod{W^3}, \quad (5.15)$$

and a ternary correction must solve

$$d\theta^{(2)} = -\frac{1}{2}[\theta^{(1)}, \theta^{(1)}]. \quad (5.16)$$

This is the universal arity-three obstruction; its value in a particular geometric theory depends on the specified colored compositions and signs.

5.1 Deforming an associative comparison

The comparison equation has an elementary instance in ordinary associative algebras. Let A, B be unital $\mathbb{k}$ -algebras, and let $f : A \rightarrow B$ be a unital algebra map. Put $R = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. On the free R -modules $A \otimes R$ and $B \otimes R$, consider products

$$a *_A b = ab + \varepsilon \alpha(a, b), \quad u *_B v = uv + \varepsilon \beta(u, v).$$

Here $\alpha : A \otimes A \rightarrow A$ and $\beta : B \otimes B \rightarrow B$ are $\mathbb{k}$ -bilinear and vanish when an input is the unit. For a bimodule M over A , define the Hochschild differential in degrees one and two by

$$\begin{aligned} (\delta h)(a, b) &= a \cdot h(b) - h(ab) + h(a) \cdot b, \\ (\delta c)(a, b, c') &= a \cdot c(b, c') - c(ab, c') + c(a, bc') - c(a, b) \cdot c'. \end{aligned}$$

The bimodule B_f has actions $a \cdot u = f(a)u$ and $u \cdot a = uf(a)$. We write δ_f for its differential. Associativity of the deformed products is equivalent to $\delta\alpha = 0$ and $\delta\beta = 0$.

Proposition 5.1 (First-order comparison obstruction). *Assume both deformed products are associative. The cochain*

$$O(a, b) = f(\alpha(a, b)) - \beta(f(a), f(b))$$

is a normalized Hochschild two-cocycle of A with values in B_f . A unital R -linear algebra map $f_\varepsilon = f + \varepsilon\varphi$ lifting f exists precisely when

$$\delta_f \varphi = O, \quad \varphi(1) = 0. \tag{5.17}$$

Thus its obstruction is $[O] \in HH^2(A, B_f)$, computed using normalized cochains. If a lift exists, its possible first-order corrections form a torsor under the normalized one-cocycles $Z^1(A, B_f)$.

Proof. Multiplicativity of f gives $\delta_f(f\alpha) = f(\delta\alpha)$ and $\delta_f(\beta(f, f)) = (\delta\beta)(f, f, f)$. Both vanish, so O is a cocycle. The coefficients of ε in $f_\varepsilon(a *_A b) = f_\varepsilon(a) *_B f_\varepsilon(b)$ give

$$f\alpha(a, b) + \varphi(ab) = \beta(f(a), f(b)) + f(a)\varphi(b) + \varphi(a)f(b).$$

This is (5.17). The unit equation is $\varphi(1) = 0$. There are no higher coefficients because $\varepsilon^2 = 0$. Two solutions differ by a normalized one-cocycle, proving the last claim. $\square$

Example 5.2 (A flat deformation without an augmentation lift). Take $A = \mathbb{k}[x]/(x^2)$, $B = \mathbb{k}$, and $f(x) = 0$. The deformation

$$A_R = R[x]/(x^2 - \varepsilon)$$

is free over R with basis $1, x$, since its defining polynomial is monic. Its first-order product has $\alpha(x, x) = 1$, while $\beta = 0$ for $B_R = R$. Consequently $O(x, x) = 1$. Every linear cochain $\varphi : A \rightarrow B$ satisfies $(\delta_f \varphi)(x, x) = 0$, since $f(x) = 0$ and $x^2 = 0$ in A . The obstruction class is nonzero. Directly, an augmentation lift would send x to εc for some $c \in \mathbb{k}$. Its square is zero, whereas the relation requires that square to equal ε .

The obstruction uses the chosen map as well as the two products. For a local field-theoretic action, its analogue must be computed in the allowed local operation complex. A primitive in a larger space of arbitrary linear maps need not define a local correction. Higher composition and overlap equations belong to the same coupled deformation problem, which is why the completed carrier is needed.

6 Representatives, gauge classes, and derived limits

Compatible representatives have an elementary limit theorem.

Theorem 6.1 (Limit of Maurer–Cartan representatives). *For an additively filtered dg Lie algebra, the map*

$$\mathrm{MC}(\mathfrak{g}) \longrightarrow \varprojlim_r \mathrm{MC}(\mathfrak{g}/F^r \mathfrak{g}) \quad (6.1)$$

is surjective if

$$\mathfrak{g}^1 \longrightarrow \varprojlim_r (\mathfrak{g}/F^r \mathfrak{g})^1$$

is surjective and $\bigcap_r F^r \mathfrak{g}^2 = 0$. It is injective if $\bigcap_r F^r \mathfrak{g}^1 = 0$. In particular, strict completeness and separation give a bijection in (6.1).

Proof. A compatible family on the right has a degree-one lift $a \in \mathfrak{g}^1$ under the stated surjectivity. Its curvature has zero image in every quotient, so degree-two separation gives $\mathrm{Curv}(a) = 0$. Degree-one separation gives uniqueness. No derived-limit term occurs because the representatives were already compatible. $\square$

The same proof applies to (4.3) under strict completeness and the additive convergence hypotheses. Vanishing of every $H^2(A_r)$ is a convenient sufficient condition for constructing a compatible dg Lie sequence recursively from the zero element in $\mathfrak{g}/F^1 \mathfrak{g}$. The mere nonemptiness of each finite solution set does not choose such a sequence.

Gauge classes obey a homotopical limit theorem instead. Assume for the remainder of this paragraph that $\mathfrak{g}$ is strictly complete and separated, and put

$$X_r = \mathrm{MC}_\bullet(\mathfrak{g}/F^r \mathfrak{g}), \quad X = \mathrm{MC}_\bullet(\mathfrak{g}) = \varprojlim_r X_r, \quad (6.2)$$

where the tensor products in X are completed. The X_r are Kan complexes, and the maps $X_{r+1} \rightarrow X_r$ are Kan fibrations. For a compatible basepoint $a = (a_r)$, the Milnor sequence for a tower of fibrations gives an exact sequence of pointed sets

$$* \longrightarrow \varprojlim_r^1 \pi_1(X_r, a_r) \longrightarrow \pi_0(X) \longrightarrow \varprojlim_r \pi_0(X_r) \longrightarrow *. \quad (6.3)$$

Here $\varprojlim^1$ is the nonabelian orbit-set construction for a tower of groups. The group $\pi_1(X_r, a_r)$ is the derived stabilizer; in dg Lie terms it is $H^0((\mathfrak{g}/F^r)^{a_r})$ with its Campbell–Hausdorff group law. Thus compatible finite-stage gauge classes have a global class, but they determine it uniquely only when the pointed set

$$\varprojlim_r^1 \pi_1(X_r, a_r) \quad (6.4)$$

is trivial. A Mittag–Leffler stabilizer tower, in particular a tower with surjective transition maps, is a standard sufficient condition. The exact sequence (6.3) is the degree-zero case of the homotopy-limit sequence for towers of fibrations [4].

There is a related, but distinct, $\varprojlim^1$ for compatible primitives. For a countable inverse system of cochain complexes (C_r, p_r) , let

$$s : \prod_r C_r \longrightarrow \prod_r C_r, \quad s((c_r)) = (p_r(c_{r+1})).$$

Then

$$\mathbf{R}\varprojlim C_r \simeq \mathrm{Cone} \left(1 - s : \prod_r C_r \longrightarrow \prod_r C_r \right) [-1] \quad (6.5)$$

and there is a natural exact sequence

$$0 \longrightarrow \varprojlim_r^1 H^{j-1}(C_r) \longrightarrow H^j(\mathbf{R}\varprojlim_r C_r) \longrightarrow \varprojlim_r H^j(C_r) \longrightarrow 0. \quad (6.6)$$

Indeed, products are exact in vector spaces, so $H^j(\prod_r C_r) = \prod_r H^j(C_r)$. The long exact cohomology sequence of (6.5) identifies its two adjacent terms with the cokernel of $1 - s$ on $\prod_r H^{j-1}(C_r)$ and the kernel of $1 - s$ on $\prod_r H^j(C_r)$, which are respectively $\varprojlim_r^1 H^{j-1}(C_r)$ and $\varprojlim_r H^j(C_r)$. When the transition maps are degreewise surjective, the ordinary inverse limit represents (6.5). This does not force the left term of (6.6) to vanish, because the induced cohomology maps need not be Mittag-Leffler.

Under this degreewise-surjectivity hypothesis, suppose compatible cocycles $c_r \in Z^j(C_r)$ are exact at every stage. Choose b_r with $db_r = -c_r$. Then

$$\delta_r = b_r - p_r(b_{r+1}) \in Z^{j-1}(C_r) \quad (6.7)$$

represents the residual class in $\varprojlim_r^1 H^{j-1}(C_r)$. Compatible primitives exist precisely when the single class of (c_r) in $H^j(\mathbf{R}\varprojlim_r C_r)$ vanishes; after stagewise vanishing, this is equivalent to the vanishing of the residual (6.7). There is no natural decomposition of this class as an element of

$$\varprojlim_r H^j(C_r) \oplus \varprojlim_r^1 H^{j-1}(C_r)$$

without a chosen splitting of (6.6). For ordinary Maurer–Cartan curvature $j = 2$, so the residual primitive group is $\varprojlim_r^1 H^1$, not $\varprojlim_r^1 H^0$. Without a quasi-isomorphism $\varprojlim_r C_r \rightarrow \mathbf{R}\varprojlim_r C_r$, derived vanishing controls a homotopy-coherent primitive and need not produce a strictly compatible one.

7 Sharp failures

Example 7.1 (Separated but incomplete). Let $\mathfrak{a}^1 = t\mathbb{k}[t]$, with zero differential and bracket, and $F^r \mathfrak{a}^1 = t^r \mathbb{k}[t]$. The compatible elements

$$a_r = t + t^2 + \cdots + t^{r-1} \in \mathfrak{a}/F^r \mathfrak{a}$$

are Maurer–Cartan. No element of $t\mathbb{k}[t]$ represents their limit $\sum_{j \geq 1} t^j$. Separation without completeness does not give surjectivity in (6.1).

Example 7.2 (Complete quotients with hidden curvature). Use the dg Lie algebra of Example 1.1, but put

$$F^1 \mathfrak{g} = \mathfrak{g}, \quad F^r \mathfrak{g} = \mathbb{k}z \quad (r \geq 2).$$

Every quotient sees x as Maurer–Cartan, while $\text{Curv}(x) = z \neq 0$. The curvature lies in $\bigcap_r F^r \mathfrak{g}^2$. Degree-two separation in Theorem 6.1 is therefore necessary.

Example 7.3 (A non-Mittag-Leffler cohomology tower). Let

$$C_r = \left[\mathbb{k}[x] \xrightarrow{\text{mod } x^r} \mathbb{k}[x]/(x^r) \right]$$

in degrees 0 and 1, with transition maps the identity in degree 0 and the quotient in degree 1. These maps are degreewise surjective, but

$$H^0(C_r) = (x^r), \quad H^1(C_r) = 0,$$

and the maps on H^0 are the proper inclusions $(x^{r+1}) \hookrightarrow (x^r)$. Hence

$$\varprojlim_r C_r = [\mathbb{k}[x] \hookrightarrow \mathbb{k}[[x]]], \quad H^1(\varprojlim_r C_r) = \mathbb{k}[[x]]/\mathbb{k}[x] \cong \varprojlim_r^1(x^r). \quad (7.1)$$

The compatible degree-one cocycles

$$1 + x + \cdots + x^{r-1} \pmod{x^r}$$

are exact separately, but their formal-series limit has no polynomial primitive.

Example 7.4 (Finite gauge classes do not determine the total class). Regard

$$\mathfrak{a}^0 = \mathbb{k}[x], \quad \mathfrak{a}^1 = \mathbb{k}[[x]], \quad d(f) = f, \quad [-, -] = 0$$

as an abelian dg Lie algebra, and set, for $r \geq 2$,

$$F^r \mathfrak{a}^0 = 0, \quad F^r \mathfrak{a}^1 = x^{r-1} \mathbb{k}[[x]], \quad F^1 \mathfrak{a} = \mathfrak{a}.$$

It is strictly complete and separated. In every finite quotient the map $d : \mathfrak{a}^0 \rightarrow \mathfrak{a}^1/F^r$ is surjective, so there is one gauge class. Globally,

$$\mathrm{MC}(\mathfrak{a})/\mathrm{gauge} = H^1(\mathfrak{a}) = \mathbb{k}[[x]]/\mathbb{k}[x]. \quad (7.2)$$

This is precisely the $\varprojlim^1$ ambiguity in (6.3).

Example 7.5 (Stagewise solvability in a curved algebra). Let $K^1 = \mathbb{k}[t]$, $K^2 = \mathbb{k}[[t]]$, let $d : K^1 \rightarrow K^2$ be inclusion, and let the bracket vanish. Give this curved abelian dg Lie algebra the curvature

$$\omega = \sum_{j \geq 0} t^j$$

and filtration

$$F^1 K = K, \quad F^r K^1 = 0, \quad F^r K^2 = t^{r-1} \mathbb{k}[[t]] \quad (r \geq 2).$$

The filtered complex is strictly complete. In $K/F^r K$, the curved Maurer–Cartan equation $\omega + da = 0$ has the solution

$$a_r = -(1 + t + \cdots + t^{r-2}).$$

There is no compatible family of solutions: compatibility would force one polynomial to agree with $-\omega$ to every order. Thus the nonemptiness of every finite solution set does not replace a compatible sequence. More fundamentally, a nullary bracket in a curved L_∞ algebra makes q_1^2 contain a bracket with q_0 ; the cohomology group in (4.7) is then unavailable unless additional filtration hypotheses make the associated-graded unary operator square to zero. The main theorems are therefore flat.

Example 7.6 (Completeness without additive convergence). Let $L^0 = \mathbb{k}x$, $L^1 = \mathbb{k}z$, and define shifted brackets by

$$q_n(x^n) = n!z \quad (n \geq 2),$$

with every bracket containing z equal to zero. All L_∞ identities hold because every composite bracket vanishes. The discrete filtration $F^1 L = L$, $F^2 L = 0$ is complete and separated, but it is not additive for the higher brackets. The series $R(x) = \sum_{n \geq 2} z$ does not converge. In the same way, $\mathfrak{sl}_2(\mathbb{k})[[t]]$ with $F^1 \mathfrak{g} = \mathfrak{g}$ and $F^r \mathfrak{g} = t^{r-1} \mathfrak{g}$ for $r \geq 2$ is complete but not additively filtered: a constant adjoint action need not converge in the t -adic topology. Completeness is useful only after the operations have been shown to move terms deeper.

8 Local actions and their total obstruction

A mixed holomorphic–topological theory has holomorphic dependence in one set of directions and topological dependence in the transverse directions. Its bulk observables come from specified fields, an action, and a quantization. Boundary conditions and boundary-changing operators give a separate boundary theory. A comparison must act on those operators and preserve their compositions, including homotopies on overlaps. Proposition 5.1 computes the first extra equation in the ordinary associative case. For the full local theory, the analogous obstruction tower requires the following data:

1. a semicosimplicial diagram of complete local deformation dg Lie algebras with strict filtration-preserving cofaces, or a separately specified complete filtered L_∞ homotopy-limit carrier;
2. a positive additive filtration and completed tensor products for all infinite operation spaces;
3. full filtered L_∞ comparison morphisms, including their higher Taylor coefficients;
4. for an operadic model, a reduced conilpotent colored cooperad and a specified quasi-free resolution, together with its continuous colored endomorphism object;
5. a flat total carrier with zero nullary bracket. Additional BV or central terms must be incorporated into a square-zero unary operator and compatible higher brackets. A genuinely curved carrier is covered only after twisting by a specified Maurer–Cartan basepoint makes its nullary bracket vanish and preserves the additive filtration.

An arbitrary chain of deformation complexes is not a totalization: its diagram, homotopy coherences, cross-operations, and total differential must be given. Likewise, projected obstruction classes are consequences of the total class, not coordinates of it unless joint detection such as (3.10) is proved.

For the dg Lie alternative in item 1, Theorem 3.4 and Theorem 6.1 give the filtered-and-totalized obstruction tower. For a separately specified complete flat L_∞ carrier, Proposition 4.1 gives its successive obstruction classes and the same completeness argument gives the limit of compatible representatives. These results do not by themselves construct a particular mixed holomorphic–topological colored carrier or evaluate its arity-three class; that value is the explicit composition (5.15) in the chosen chain-level model.

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